Sourav Yadav

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Summary

Final-year CSE student at IIIT Manipur building ML models and LLM-powered features end to end — scikit-learn/XGBoost models on large, messy datasets, RAG pipelines with embeddings, and FastAPI model APIs backed by SQL. Strong in Python, Pandas, and NumPy, with a WABI 2026 co-authored research publication in applied ML.

Education

Indian Institute of Information Technology (IIIT) Senapati, Manipur — Aug 2023 – Present

- B.Tech in Computer Science and Engineering | CPI: 8.85 (6th Sem)

Experience

Research Intern, National Institute of Technology Kurukshetra — Nov 2025 – Apr 2026

Cleaned and analyzed a large, messy real-world dataset of 18,104+ genomes, engineering features and validating data quality to build a predictive model surfacing biologically meaningful patterns.
Evaluated results with standard statistical metrics (**F1-macro = 0.79**, **AUROC = 0.84**) and presented findings in a paper accepted at **WABI 2026**.
Tools: Python, PyTorch, Statistical Analysis, NumPy

Publications

- Co-authored the paper "*Biologically Grounded Explainable Antimicrobial Resistance Prediction Using Knowledge Graphs*", accepted for presentation and publication at **WABI 2026** (26th International Conference on Algorithms for Bioinformatics).

Projects

AI-Powered Fraud Detection System — github-repo | Demo-link

Cleaned and analyzed 284K+ transactions, engineering risk features and training an ensemble (Random Forest, XGBoost, Neural Network) for a highly imbalanced problem (class weighting, stratified splits), achieving **AUC-ROC $\approx$ 0.98**.
Used SHAP for explainable, per-user risk profiles and served predictions through a **PostgreSQL**-backed **FastAPI** model API.
Tools: Python, Pandas, NumPy, SQL, PostgreSQL, Scikit-learn, XGBoost, SHAP, FastAPI

Webpage Chat Assistant (Chrome Extension) — github-repo

Built a **RAG pipeline** (chunking, embeddings, retrieval, query rewriting) over an **LLM API** so answers are grounded in live webpage content.
Served it through a **FastAPI** backend with per-session tracking; diagnosed a production reliability issue and fixed it with TTL/LRU eviction and capped context sizing.
Tools: Python, FastAPI, LangChain, Groq API, Pydantic

Trip Planning Agent (LLM Agent, FastAPI + PostgreSQL) — github-repo | Demo-link

Built and deployed an **LLM tool-calling agent** (LangGraph) over 7 tools behind a **FastAPI** backend, with **PostgreSQL**-backed conversation state and human approval before any booking step.
Fixed post-deployment failures: stale DB connections (health-checked connection pool), LLM rate-limit breaches (dynamic context trimming), and silent errors (added structured logging).
Tools: Python, LangGraph, FastAPI, PostgreSQL, React, Git

Technical Skills

Languages: Python, SQL, C++, C, JavaScript
Data & ML: Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, TensorFlow, SHAP, Feature Engineering, Model Evaluation
LLM / GenAI: LLM APIs, Embeddings, RAG Pipelines, Prompt Engineering, LangChain, LangGraph
Backend & Databases: FastAPI, REST APIs, Pydantic, PostgreSQL, MongoDB, SQLite, SQLAlchemy
Tools & Deployment: Git, Docker, Jupyter Notebook, Render, Vercel